

Water Electrolysis

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Split water and see a 2:1 ratio

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 4 — Water Electrolysis and Gas Collection

Hook: Split water, collect gases, and see a 2:1 stoichiometric ratio.

Learning outcomes

- Build a gas collection apparatus by water displacement
- Run water electrolysis safely with a non-chloride electrolyte
- Measure H₂ and O₂ volumes and compare to 2:1 ratio
- Connect electrolysis to hydrogen as an energy storage concept
- **Start** electrolytic rust-removal cells for Session 5 reveal

Session plan (90 min)

Time	Activity	Reference
0–10 min	Hook: inverted water-filled tubes — which gas appears faster?	lecture.md
10–25 min	Water splitting and stoichiometry	lecture.md
25–40 min	Build gas collection setup	experiment.md — Part A
40–60 min	Run electrolysis; record volumes over time	experiment.md — Part B
60–70 min	Discuss ratio; optional instructor H ₂ pop test	experiment.md — Part C
70–80 min	Brief energy storage discussion	lecture.md
80–90 min	Start derusting cells for Session 5	experiment.md — Part E

Key reactions

Overall: $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$

Cathode (–): $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$

Anode (+): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ (acidic media)

or $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$ (basic media)

Status

- [] Graphite electrodes sourced
- [] Gas collection apparatus pilot-tested
- [] **No saltwater** — baking soda or Na₂SO₄ only
- [] Pop test protocol reviewed
- [] Rusty nails + washing soda ready for Part E

Session 4 — Lecture: Water Electrolysis and Stoichiometry

Target duration: ~20 minutes (+ energy discussion at end)

Instructors: half-reactions, why no NaCl, 2:1 ratio errors, derusting handoff — instructor.md.

Opening hook (0–10 min)

Show inverted test tubes filled with water in a tub.

"One electrode makes hydrogen. One makes oxygen. Which side will fill twice as fast? Why?"

Core concepts (10–25 min)

1. Electrolysis of water

- Pure water conducts poorly → add **baking soda** or **sodium sulfate** (NOT NaCl)
- Power supply forces non-spontaneous reaction: water → hydrogen + oxygen

2. Half-reactions (simplified)

Cathode (–): water reduced → **hydrogen gas**

Anode (+): water oxidized → **oxygen gas**

3. Stoichiometry — the 2:1 ratio

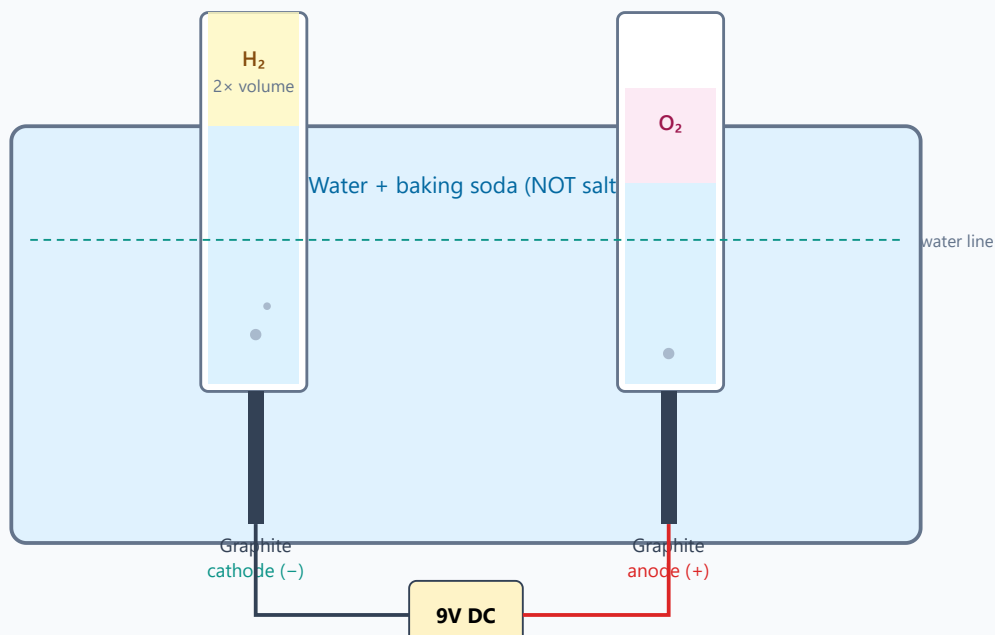
From overall equation: $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

- **2 volumes H_2 : 1 volume O_2** (same T and P)
- Avogadro: 2 moles H_2 : 1 mole O_2

4. Gas collection by displacement

- Tubes start **100% full of water** — no air bubble
- Gas pushes water out as it forms
- Measure **volume of gas space** at intervals

Session 4 - Water electrolysis + gas collection



Fill tubes completely with water, invert underwater - measure gas volume over time

Expected ratio $V(\text{H}_2) : V(\text{O}_2) \approx 2 : 1$

5. Why observed ratio may differ from 2:1

- O_2 slightly more soluble in water than H_2
- Small bubbles escape before collection
- Leaks, uneven electrode areas
- Still a strong qualitative and semi-quantitative demo

Energy discussion (75–85 min)

Hydrogen as energy storage

1. **Excess renewable electricity** (solar/wind) $\rightarrow$ electrolysis $\rightarrow \text{H}_2$
2. Store H_2 (compressed, or other methods — brief mention)
3. Later: **fuel cell** converts $\text{H}_2 + \text{O}_2$ back to electricity + water

Key idea: Hydrogen is an **energy carrier**, not a primary energy source — energy is lost at each step.

Efficiency chain (qualitative)

Solar $\rightarrow$ electricity $\rightarrow \text{H}_2 \rightarrow$ storage $\rightarrow$ fuel cell $\rightarrow$ electricity $\rightarrow$ motor/light
(Each arrow loses some energy as heat)

Preview Session 5 (85–90 min)

"We used electricity to make hydrogen. Overnight we'll use electricity to undo rust on iron. Tomorrow: reveal the results, then connect corrosion, galvanic cells, and sacrificial protection."

Start derusting cells per experiment.md Part E before students leave.

Visual aids

- ☐ Balanced equation with mole/volume ratio highlighted
 - ☐ Photo of setup with gas volumes marked
 - ☐ Simple energy diagram: renewable → H₂ → (optional fuel cell concept)
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Vocabulary

- ☐ Electrolysis (applied to water)
 - ☐ Stoichiometry / mole ratio
 - ☐ Water displacement
 - ☐ Energy carrier vs. energy source
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Safety talking points (brief in lecture, detailed in experiment)

- **Never use saltwater** — chlorine gas risk
- No open flames near apparatus
- Pop test: instructor only, tiny H₂ volume

Session 4 — Instructor Notes: Water Electrolysis Deep Dive

Audience: instructors and TAs only. Pair with lecture.md and experiment.md.

Why this session exists

Students have plated metals. Now electricity splits **molecules** of water into elemental gases, and stoichiometry becomes visible as a **2:1 volume ratio**. This also introduces hydrogen as an energy carrier and sets up overnight electrolytic rust removal for Session 5.

Atomic / molecular foundations

Water's bonding (instructor talking points)

- H_2O : oxygen shares electrons with two hydrogens (polar covalent bonds).
- Oxygen is more electronegative → partial negative charge on O, partial positive on H.
- Liquid water also has extensive hydrogen bonding (affects boiling point, solvation) — optional aside.
- Pure water has very few ions (autoionization): $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$ with tiny $[\text{H}^+][\text{OH}^-] = 10^{-14}$ at 25 °C. That is why **pure water is a poor conductor**.

Why we add baking soda or Na_2SO_4

Electrolysis needs mobile ions. Baking soda (NaHCO_3) or sodium sulfate (Na_2SO_4) dissolves into ions that carry current. They are **supporting electrolytes** — they enable conduction without (ideally) being the main reactants consumed.

Critical safety contrast:

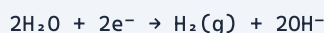
Electrolyte	OK for classroom gas electrolysis?	Why
Baking soda / NaHCO_3	Yes	No chlorine evolution
Sodium sulfate	Yes	Inert supporting electrolyte
Table salt NaCl	NO	Chloride oxidizes → Cl_2 gas (toxic)

If anyone smells chlorine or sees greenish gas character — **stop immediately**, ventilate, replace electrolyte.

Half-reactions — what to teach vs what is “true”

Classroom-friendly version

Cathode (–):



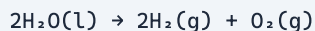
(or in acid: $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$)

Anode (+):



(or in base: $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$)

Overall:



Why the 2:1 ratio appears

Oxygen half-reaction involves **4 electrons** per O_2 ; hydrogen involves **2 electrons** per H_2 . Balancing electrons requires twice as many H_2 molecules as O_2 molecules:



At the same T and P, gas volume $\propto$ moles (Avogadro / ideal gas). So volumes follow mole ratio.

Electron bookkeeping (Faraday connection)

Same idea as Session 3: charge passed determines moles of e^- , which determines moles of gas — but today students measure **volumes**, not mass.

Thermodynamics (instructor background)

Water splitting is **non-spontaneous** under standard conditions. Rough minimum theoretical voltage for water electrolysis is $\sim 1.23 \text{ V}$ (thermodynamic), but real cells need more due to overpotentials and resistance — classroom 9 V packs are plenty. Students need only: *"We must push with a battery; water does not split by itself."*

Gas collection by water displacement

Why tubes start full of water

Any air bubble at the start ruins the stoichiometric comparison. Gas collected = displaced water volume.

Identifying H_2 vs O_2 without a pop test

Clue	H_2 side	O_2 side
Rate	Faster ($2\times$ volume)	Slower
Electrode	Cathode (–)	Anode (+)
Optional	Pop test (instructor)	Relights glowing splint (advanced / careful)

Default classroom ID: **faster = H_2 = cathode**.

Why ratios drift from 2.0

Cause	Effect
O ₂ more soluble than H ₂	O ₂ volume low → ratio high
Leaks / escaped bubbles	Either side low
Unequal electrode area / current distribution	Skewed rates
Air left in tube at t = 0	Systematic error
Electrode oxidation / side reactions	Extra gases or lost current

Accept **1.6–2.2** as a successful classroom result; discuss systematics rather than demanding perfection.

Pop test protocol (instructor only)

1. Collect a **tiny** volume of H₂ in a separate small tube.
 2. Move away from the main tub and from faces.
 3. Brief flame → soft “pop.”
 4. Never ignite sealed containers or mixed H₂/O₂.
 5. Skip entirely if venue rules forbid open flame.
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Hydrogen as energy storage (end-of-session talk)

Frame carefully:

1. Renewables make electricity intermittently.
2. Extra electricity can electrolyze water → H₂.
3. H₂ can be stored and later oxidized in a **fuel cell** → electricity + water.
4. H₂ is an **energy carrier**, not a source — the energy came from the electricity (ultimately the sun/wind).
5. Round-trip efficiency < 100% always (heat losses).

You are **not** running a fuel cell demo this week (Session 5 uses rust removal instead). Keep fuel cells as a conceptual bridge only.

Part E — Start electrolytic rust removal (handoff to Session 5)

Correct polarity (students reverse this)

Electrode	Connection	Role
Rusty iron (object to clean)	Battery – (cathode)	Reduction / cleaning
Sacrificial steel or graphite	Battery + (anode)	Oxidizes / sacrifices
Electrolyte	Washing soda or baking soda	Not NaCl

What is happening chemically (simplified)

Rust is largely $\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ / FeOOH mixtures. At the cathode, electrochemical reduction and associated chemistry loosen / reduce oxide layers toward metallic iron; H_2 evolution often accompanies the process. The anode corrodes or evolves O_2 depending on material.

Teaching link: same “electricity drives non-spontaneous change” idea as water electrolysis — applied to undoing corrosion.

Overnight logistics

- Label cells; photograph “before.”
- Keep a no-power control nail in water.
- Confirm venue policy for leaving batteries connected overnight (prep room preferred).

Misconceptions

Misconception	Correction
“Bubbles are steam.”	They are H_2 and O_2 gases , not water vapor from boiling.
“Salt makes electrolysis better.”	Salt makes chlorine — forbidden here.
“Ratio must be exactly 2.000.”	Classroom data have solubility and collection errors.
“Hydrogen is a fuel like oil.”	It’s a carrier ; energy was put in by electrolysis.

Board plan

1. Write overall $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$; circle 2:1.
2. Assign cathode/anode gases with signs.
3. Sketch displacement tubes; emphasize “no air at start.”
4. Safety slide: no NaCl .
5. Preview energy carrier concept + start derusting cells before dismissal.

Optional enrichment

- Pourbaix diagrams for $\text{Fe-H}_2\text{O}$ (why rust forms / removes — advanced).
- Overpotential differences for H_2 vs O_2 on graphite.
- Industrial alkaline electrolyzers vs PEM electrolyzers (names only).

Session 4 — Preparation

Before class (instructor)

- [] **Pilot full setup** — confirm 2:1 ratio visible within session time
- [] Pre-cut graphite electrodes to useful lengths
- [] Mix electrolyte batch: water + baking soda (label "NO SALT")
- [] Test tubes: practice inversion technique — no trapped air
- [] Review pop test safety; decide if demo will run (weather, room rules)
- [] Optional: test solar panel output with same cell
- [] Stage rusty nails, washing soda, sacrificial anodes, and spare jars for **Part E** (end of class)
- [] Confirm overnight power policy for derusting cells (prep room vs. disconnect)

Day-of setup

- [] Pre-fill tub at each station (or central fill station)
- [] Pre-fill inverted tubes underwater as demo for students
- [] Large sign: "**No salt in electrolysis**"
- [] Separate instructor area for pop test marked off
- [] Part E station: rusty nails, washing soda, steel/graphite anodes, labels, camera

Common failure modes (from pilot)

Issue	Prevention
Air in tubes	Demo inversion step slowly
Weak bubble rate	Fresh baking soda; clean graphite
Students add salt "to speed up"	Pre-brief; labeled solutions

Open questions / notes

- Electrode material used:
- Best tube type for measurement:
- Pop test yes/no for this venue: